

- 4 (a)** A study was carried out to investigate natural selection in the plant evening primrose, *Oenothera biennis*.

Physiological changes associated with resistance to grazing by herbivorous insects were measured over 5 years in 16 experimental populations of evening primrose. For 8 of the experimental populations the plants were regularly sprayed with chemical insecticide, and the other 8 experimental populations were not sprayed.

In each of the 5 years some older plants died after producing seeds and some new plants grew from seed.

Observations after the study were:

- sprayed populations:
 - The length of time that the plants produced flowers (flowering period) became longer.
 - Flowering started earlier in each generation.
 - The mean concentration of natural insect-deterring chemicals in the plants was relatively low.
- non-sprayed populations:
 - The flowering period remained the same over the 5 years.
 - The mean concentration of natural insect-deterring chemicals in the plants increased.

Analysis showed that genetic differences were responsible for the differences in flowering time and concentration of natural insect-deterring chemicals in the plants. The researchers concluded that natural selection was acting on both groups of plants.

Explain how natural selection acted in the **non-sprayed** populations to cause the mean concentration of natural insect-deterring chemicals to increase.

[4]

- (b) (i) Identify the type of natural selection that caused an increase in the mean concentration of insect-detering chemicals in the non-sprayed populations.

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- (ii) Identify the type of natural selection that caused the flowering period to remain the same in the non-sprayed populations.

.....[1]

- (c) The same trends in results were recorded in all of the non-sprayed populations.

Explain how this supports the researchers' conclusion that natural selection caused the trends and **not** genetic drift.

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[Total: 8]